

Capstone Project Proposal

Project Category & Title:

Project S: Campus appointment scheduler

Problem & Users:

Working out availability for advising appointments and office hours can be clunky, especially with multiple students all trying to schedule appointments in a small timeframe. Scheduling through methods like email communication can lead to overly restrictive options when trying to set up appointments with multiple students. This application will provide a portal for professors to dictate their availability, and for students to request appointments within that availability, optimizing the options for students and streamlining the multiple appointment making process for professors.

The application will serve two primary user types: professors/advisors with strict availability times and students who require appointments.

Core Features:

Must-have:

- User login/authentication with professor/student roles
- View Professor Availability
- Book Appointment
- Appointment management and conflict prevention

Nice-to-have:

- Email Notifications
- Interactive Calendar View

Architecture & Tech Stack:

Architecture:

React frontend → FastAPI API backend → PostgreSQL database

Tech Stack:

Languages: Python, JavaScript, SQL

Frontend: React

Backend: FastAPI

Database: PostgreSQL

ORM: SQLAlchemy

Authentication: Google OIDC

External Integration: Google Calendar API

Version Control: Git/Github